

# Congenital Thrombotic Thrombocytopenic Purpura (Upshaw– Schulman Syndrome): A Comprehensive Disease Characteristics Report

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## Summary

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Congenital thrombotic thrombocytopenic purpura (cTTP), also known as **Upshaw–Schulman syndrome (USS)** or **hereditary TTP (hTTP)**, is an ultra-rare, life-threatening thrombotic microangiopathy caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in the *ADAMTS13* gene** on chromosome 9q34.2 (OMIM #274150; gene OMIM \*604134). The genetic defect produces severe, lifelong deficiency (typically <10%, often <5–6% activity) of ADAMTS13, the plasma metalloprotease that cleaves von Willebrand factor (VWF). Inheritance is **autosomal recessive**.

The central pathophysiology is elegantly linear: absent ADAMTS13 protease activity allows **ultra-large VWF (UL-VWF) multimers** to persist in circulation. Under high shear stress in the microvasculature, these multimers spontaneously bind and aggregate platelets, generating disseminated platelet-rich microthrombi. This produces the clinical triad of **consumptive thrombocytopenia, microangiopathic hemolytic anemia (MAHA), and ischemic end-organ injury** — most prominently affecting the brain, kidney, and heart. Because ADAMTS13 activity is already at baseline near-zero, additional physiological stressors that raise VWF or shear (birth, pregnancy, infection, inflammation) act as "second hits" that precipitate acute episodes, explaining the marked (~78%) female predominance and the frequent neonatal and pregnancy-associated presentations.

cTTP is a treatable disease when recognized: the therapeutic principle is **ADAMTS13 replacement**. Historically this was achieved with fresh frozen plasma (FFP) or plasma-derived factor VIII/VWF concentrates that contain ADAMTS13. The field has now been transformed by **recombinant ADAMTS13 (apadamtase alfa; Adzynma/TAK-755)**, which in a pivotal phase 3

crossover trial (NCT03393975) prevented essentially all acute TTP events. This report synthesizes ten confirmed findings across 31 reviewed papers to populate a complete 15-section disease knowledge base entry.

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## Key Findings

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### Finding 1 — cTTP is caused by biallelic *ADAMTS13* mutations (autosomal recessive)

Congenital TTP results from **homozygous or compound heterozygous disease-causing variants in *ADAMTS13*** (chromosome 9q34.2; OMIM #274150; gene OMIM \*604134). The consequence is severe deficiency of the VWF-cleaving metalloprotease, with activity typically <10% and often <5–6%. More than **200 distinct pathogenic variants** are distributed across the gene, with **missense variants predominating**, alongside frameshift, nonsense, and splice-site variants. Inheritance is unambiguously **autosomal recessive**, meaning both alleles must be affected for disease to manifest.

Foundational evidence comes from Levy and colleagues, who established that *"homozygous or compound heterozygous mutations of *ADAMTS13* are responsible for recessively inherited TTP"* (PMID: 12393505). More recent reviews confirm cTTP is *"a rare genetic disorder caused by mutations in the *ADAMTS13* gene that leads to decreased or absent production of the plasma von Willebrand factor (VWF)-cleaving metalloprotease *ADAMTS13*"* (PMID: 37895305).

### Finding 2 — Pathophysiology: *ADAMTS13* deficiency → uncleaved ultra-large VWF → shear-dependent microthrombosis

*ADAMTS13* is a ~190 kDa **plasma metalloprotease produced mainly by hepatic stellate cells**. Its physiological function is to cleave the **Tyr1605–Met1606 scissile bond in the VWF A2 domain**. Cleavage requires tension/shear-induced unfolding of the A2 domain and allosteric activation of *ADAMTS13* through binding of its distal T8–CUB domains to the VWF D4 domain. When *ADAMTS13* is deficient, ultra-large VWF multimers accumulate; these spontaneously bind platelets under high shear, producing **disseminated microvascular platelet thrombi**, consumptive thrombocytopenia, and microangiopathic hemolytic anemia (red cells are mechanically sheared as they pass through partially occluded microvessels).

Crawley and Scully described that *"ADAMTS13, a 190-kD plasma protease originating primarily in hepatic stellate cells, prevents microvascular thrombosis by cleaving von Willebrand factor when the substrate is conformationally unfolded by high levels of shear stress"* (PMID: 19180123). The precise cleavage site was defined as the *"Cleavage of the Tyr(1605)-Met(1606) scissile bond in the VWF A2 domain"* (PMID: 17146059).

### Finding 3 — Recombinant ADAMTS13 (apadamtase alfa) prevents acute events in phase 3 trial

The pivotal **phase 3 open-label crossover trial** (NCT03393975; Scully et al., NEJM 2024) enrolled 48 patients randomized to **recombinant ADAMTS13 (40 IU/kg IV)** versus standard plasma-based prophylaxis. The results were striking: **zero acute TTP events** occurred during rADAMTS13 prophylaxis versus 1 during standard therapy (mean annualized event rate 0.05). The annualized rate of thrombocytopenia manifestations was 0.74 with rADAMTS13 versus 1.73 with standard therapy. Adverse events occurred in 71% versus 84% of patients respectively, and **no anti-ADAMTS13 antibodies** developed. The earlier first-in-human phase 1 study (BAX 930) demonstrated dose-proportional pharmacokinetics and good tolerability.

The primary efficacy result was reported as: *"No acute TTP event occurred during prophylaxis with recombinant ADAMTS13, whereas 1 patient had an acute TTP event during prophylaxis with standard therapy (mean annualized event rate, 0.05)"* (PMID: 38692292). The first-in-human safety was established: *"BAX 930 was well tolerated, no serious adverse events occurred, and no anti-ADAMTS-13 antibodies were observed"* (PMID: 28912376).

### Finding 4 — c.4143dupA (p.Glu1382Argfs) is a Northern/Central European founder mutation

The **c.4143dupA (4143insA) frameshift variant in ADAMTS13 exon 29** shows striking geographic concentration in Northern/Central Europe. It has been identified in families from Germany, Norway, Sweden, Poland, the Czech Republic, and Australia (German ancestry). A shared, identical intragenic haplotype across 17 polymorphic markers confirms a **common founder origin**, making it the single most frequent ADAMTS13 mutation in European cTTP cohorts.

Schneppenheim and colleagues demonstrated that *"The haplotypes linked to 4143insA were identical in all informative families"* and concluded that *"4143insA has a common genetic background and is frequent among patients with hereditary ADAMTS13 deficiency in Northern and Central European countries"* (PMID: 16807643).

## Finding 5 — Natural history: female predominance, high event rate without prophylaxis, neonatal risk

A retrospective multinational cohort (78 patients, 9 sites) documented that **78.2% of patients were female**, with 92 acute TTP events occurring in 70.5% of patients (0.145 events/person-year). Critically, **87% of acute events occurred WITHOUT prophylaxis**, and 20% of those unprotected events caused organ damage. Neonatal presentation is common: **35–50% of patients present with severe hemolysis, jaundice, and thrombocytopenia in the first days of life**. Pregnancy physiologically lowers ADAMTS13 activity, precipitating relapse, and the International Hereditary TTP Registry documents a clinically heterogeneous course with incomplete genotype-phenotype correlation.

The cohort reported: *"Eighty (87.0%) acute TTP events occurred in the absence of prophylactic treatment, of which 16 (20.0%) resulted in organ [damage]" (PMID: 42603082)*. The neonatal risk was quantified: *"The greatest risk for hTTP is in their first days after birth, when 35-50% of patients will have severe hemolysis, jaundice, and thrombocytopenia" (PMID: 38536644)*.

## Finding 6 — Adamts13-knockout mouse requires modifier background + trigger (gene-environment model)

Motto et al. (2005) demonstrated that **Adamts13-deficient mice are viable with normal survival** but exhibit prolonged VWF-mediated platelet-endothelial interactions. On the **CASA/Rk genetic background** (which has elevated plasma VWF), a subset develop spontaneous thrombocytopenia and decreased survival. Critically, **challenge with shigatoxin produces a syndrome closely resembling human TTP**. No correlation was observed between plasma VWF level and TTP severity, implying additional TTP-modifying genes beyond VWF. This is a foundational **gene-environment ("two-hit") model** of the disease.

The authors reported: *"Challenge of these mice with shigatoxin (derived from bacterial pathogens associated with the related human disease hemolytic uremic syndrome) resulted in a striking syndrome closely resembling human TTP"* and that *"no correlation was observed between plasma vWF level and severity of TTP, implying the existence of TTP-modifying genes distinct from vWF"* (PMID: 16200209).

## Finding 7 — Pregnancy in cTTP: high loss/morbidity untreated; plasma prophylaxis markedly improves outcomes

Davidesko et al. (2023) studied a cohort of 14 women with hTTP (homozygous c.3772delA) across **71 pregnancies**: 17 (24%) ended in pregnancy loss and 32 (45%) were complicated by severe obstetric morbidity (SOM). FFP-treated pregnancies had dramatically lower SOM (**28% vs 72%,  $p < 0.001$** ) and fewer preterm TTP exacerbations (**18% vs 82%,  $p < 0.001$** ). Elevated non-pregnant VWF antigen predicted SOM even among treated women (225% vs 165%,  $p = 0.047$ ), suggesting VWF antigen may be a useful biomarker. In a separate case, recombinant ADAMTS13 rescued a plasma-refractory pregnancy, leading to a live birth.

The benefit of prophylaxis was quantified: *"Treated women had decreased SOM (28% vs 72%,  $p < .001$ ) and preterm thrombotic thrombocytopenic purpura exacerbations (18% vs 82%,  $p < .001$ )" (PMID: 36889591). The recombinant rescue case reported: *"weekly injections of recombinant ADAMTS13 at a dose of 40 U per kilogram of body weight were initiated. The patient's platelet count normalized"* (PMID: 36546627).*

## Finding 8 — ADAMTS13 circulates in a closed (autoinhibited) conformation opened allosterically by VWF

Plasma ADAMTS13 circulates in a **folded/closed conformation** stabilized by an intramolecular interaction between the central **Spacer domain** and the **C-terminal CUB domains**. Binding of the distal domains to VWF D4(-CK), or to activating antibodies, extends ADAMTS13 into an **open, catalytically enhanced conformation**, increasing the metalloprotease-domain  $k_{cat}$  approximately 2-fold and exposing a cryptic epitope in the metalloprotease domain. Flexible linker regions around the metalloprotease and T2 domains mediate this conformational activation. This closed-to-open allosteric switch is central to understanding both physiology and engineered therapeutics.

Schelte et al. established that *"Plasma ADAMTS13 circulates in a folded conformation that is stabilized by an interaction between the central Spacer domain and the C-terminal CUB"* and that *"conformational extension of ADAMTS13 enhances the proteolytic function of the metalloprotease domain ( $k_{cat}$ ), rather than augmenting substrate binding ( $K_m$ )"* (PMID: 32196558). Deforche et al. identified the linker regions responsible for this flexibility (PMID: 26391536).

## Finding 9 — Diagnosis relies on ADAMTS13 activity <10% (FRETs-VWF73 gold standard)

TTP is defined by **severe ADAMTS13 activity deficiency (<10%)**. The **FRETs-VWF73 fluorogenic assay is the reference/gold-standard method**. The automated chemiluminescent immunoassay HemosIL AcuStar is faster but shows clinically relevant discrepancies versus FRETs-VWF73 (affecting diagnosis in 5/32 and follow-up in 7/51 samples; AcuStar reads systematically lower), partly because autoantibodies reduce activity more in AcuStar/ELISA than in FRETs assays. Novel fiber-optic surface plasmon resonance (FO-SPR) assays (detection limit ~6.8%, CV 7.2%) are in development. For **cTTP specifically, absence of an anti-ADAMTS13 inhibitor plus biallelic pathogenic ADAMTS13 variants confirms the congenital form** and distinguishes it from the acquired/immune form (iTTP).

Evidence documents that *"discrepancies between AcuStar and the gold standard FRETs-VWF73 have been documented in a manner that would affect diagnosis and treatment"* (PMID: 37063760) and that *"Thrombotic thrombocytopenic purpura (TTP) is characterized by severe ADAMTS-13 activity deficiency (<10%)"* (PMID: 37711907).

## Finding 10 — Residual-activity variants (e.g., R1060W) cause late-onset/pregnancy-triggered cTTP

Falter et al. (2014) reported a patient with compound heterozygous *ADAMTS13* **p.Q44X** (exon 2 premature stop) plus **p.R1060W** (exon 24 missense associated with low but measurable ADAMTS13 activity), who presented as **late-onset, pregnancy-induced cTTP** — first acute episode at age 19 during a first pregnancy, with a sibling who died during a second pregnancy. **R1060W is a recurrent variant enriched among adult/pregnancy-onset patients**. Registry data similarly show that higher residual ADAMTS13 activity correlates with later overt disease onset — a genotype-phenotype relationship of clinical importance.

The authors described *"a missense mutation in exon 24 (p.R1060W) associated with low but measurable ADAMTS13 activity"* and emphasized that *"Genetic analysis of the ADAMTS13 gene is important in TTP patients of all ages if an ADAMTS13 inhibitor has been excluded"* (PMID: 24994604).

# Comprehensive Disease Profile (15 Sections)

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## 1. Disease Information

**Overview.** Congenital thrombotic thrombocytopenic purpura is an ultra-rare, autosomal-recessive thrombotic microangiopathy caused by inherited severe deficiency of ADAMTS13. It manifests as episodic microvascular thrombosis producing thrombocytopenia, microangiopathic hemolytic anemia, and ischemic organ damage.

**Key identifiers:** | Resource | Identifier | |-----|-----| | OMIM (disease) | #274150 | | OMIM (gene) | \*604134 (ADAMTS13) | | Suggested MONDO | MONDO:0010134 (congenital TTP / Upshaw–Schulman syndrome) | | ICD-10 | D69.4 / M31.1 (thrombotic microangiopathy grouping) | | ICD-11 | 3B64.1 (thrombotic microangiopathy) | | MeSH | Purpura, Thrombotic Thrombocytopenic (D011697) | | Orphanet | ORPHA:93583 (hereditary/congenital TTP) | | Gene locus | 9q34.2 |

**Synonyms:** Upshaw–Schulman syndrome (USS); hereditary TTP (hTTP); congenital TTP (cTTP); familial TTP; ADAMTS13 deficiency, congenital.

**Information source type:** Predominantly derived from **aggregated disease-level resources** — the International Hereditary TTP Registry, multinational retrospective cohorts, and case series — rather than population-scale EHR, reflecting its ultra-rare status.

## 2. Etiology

**Primary cause:** Genetic — biallelic loss-of-function variants in *ADAMTS13* (Finding 1). This is the necessary and sufficient molecular cause of the congenital form.

**Genetic risk factors:** The causal variants themselves (>200 known pathogenic variants). Founder alleles increase population-specific risk, notably **c.4143dupA/p.Glu1382Argfs** in Northern/Central Europe (Finding 4). **Modifier genes** beyond *ADAMTS13* modulate severity — the mouse model demonstrated TTP-modifying genes distinct from VWF (Finding 6). Common *ADAMTS13* amino-acid polymorphisms (R7W, Q448E, P618A, A732V) can act as positive or negative modifiers of secretion/activity depending on context ([PMID: 16160007](#)).

**Environmental / triggering risk factors:** Pregnancy (physiologically lowers ADAMTS13 and raises VWF), the neonatal period/birth, infection and inflammation, and surgery. **Female sex** is a major demographic risk factor (~78% of patients) driven largely by pregnancy-triggered episodes (Finding 5).

**Protective factors:** No specific genetic protective alleles are established. The dominant *modifiable* protective factor is **prophylactic ADAMTS13 replacement** (plasma or recombinant), which prevented 87% of the acute events that occurred in its absence (Findings 5, 7).

**Gene-environment interactions:** The disease is a paradigm of gene-environment ("two-hit") interaction — an underlying genetic ADAMTS13 deficiency requires an environmental/physiological trigger (elevated VWF/shear) to precipitate overt disease. This is directly demonstrated in the Adamts13-knockout mouse, which develops TTP only on a high-VWF background AND after shigatoxin challenge (Finding 6).

### 3. Phenotypes

| Phenotype                               | Type            | Suggested HPO                             | Onset           | Frequency/Severity                |
|-----------------------------------------|-----------------|-------------------------------------------|-----------------|-----------------------------------|
| Thrombocytopenia                        | Lab abnormality | HP:0001873                                | Neonatal–adult  | Universal during episodes; severe |
| Microangiopathic hemolytic anemia       | Lab abnormality | HP:0001937 (schistocytosis)<br>HP:0001981 | Neonatal–adult  | Universal during episodes         |
| Neonatal jaundice/hyperbilirubinemia    | Clinical sign   | HP:0000952 (jaundice)                     | Neonatal        | 35–50%                            |
| Ischemic stroke / neurological deficits | Clinical sign   | HP:0002140                                | Childhood–adult | Common; can be presenting feature |
| Renal impairment                        | Lab/clinical    | HP:0000083                                | Variable        | Frequent, may be longstanding     |
| Fatigue / purpura / bleeding            | Symptom/sign    | HP:0000978 (purpura)                      | Variable        | Common                            |
| Cardiac ischemia                        | Clinical sign   | HP:0001681                                | Adult           | Less common but serious           |

**Characteristics.** Onset ranges from **neonatal (35–50%)** through childhood to adult/pregnancy-triggered late onset. Severity is **variable** and partly genotype-dependent — residual-activity alleles (R1060W) associate with milder, later-onset disease (Finding 10). The course is

characteristically **episodic/relapsing**, punctuated by triggers. **Quality-of-life impact** is substantial: recurrent hospitalizations, need for regular infusions, pregnancy loss and obstetric morbidity, and risk of permanent stroke- or renal-related disability.

#### 4. Genetic/Molecular Information

**Causal gene:** *ADAMTS13* (HGNC:1366; gene OMIM \*604134; 9q34.2), encoding the VWF-cleaving metalloprotease.

**Pathogenic variants:** >200 distinct variants distributed across all domains; **missense predominate**, with frameshift, nonsense, and splice-site variants also common (Finding 1). Classification per ACMG/AMP spans pathogenic and likely pathogenic; residual-activity missense alleles may be VUS pending functional data. Representative variants: **c.4143dupA (p.Glu1382Argfs)** — European founder frameshift; **p.R1060W** — residual-activity missense enriched in adult/pregnancy onset; **p.Q44X** — nonsense; **c.3772delA** and **c.721delG (p.Gly241fs)** — frameshift; **p.R1336W** and **p.P618A** — activity-reducing missense. Variants are **germline** (not somatic). Functional consequence is predominantly **loss of function** (impaired secretion and/or catalytic activity).

**Modifier genes:** Motto et al. established the existence of TTP-modifying genes distinct from VWF (Finding 6). Intragenic *ADAMTS13* polymorphisms modulate secretion/activity ([PMID: 16160007](#)).

**Epigenetic information / chromosomal abnormalities:** No disease-defining epigenetic changes or large-scale chromosomal abnormalities are established for cTTP; it is a single-gene monogenic disorder.

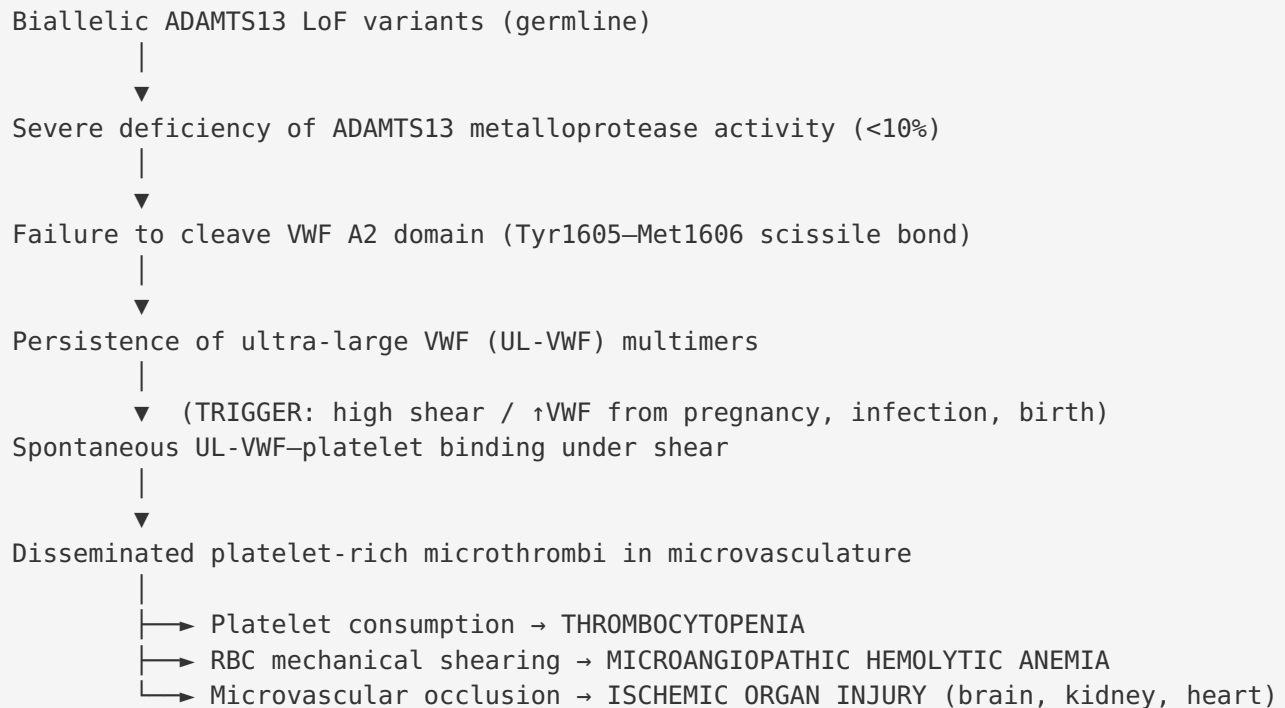
#### 5. Environmental Information

**Environmental / lifestyle factors:** cTTP is not caused by environmental toxins, radiation, or lifestyle. However, physiological/environmental **triggers** unmask disease: pregnancy, birth, infection, inflammation, and surgery.

**Infectious agents:** No pathogen causes cTTP. Notably, **shigatoxin** (from Shiga-toxin-producing bacteria, classically associated with HUS) triggers a TTP-like syndrome in genetically susceptible *ADAMTS13*-deficient mice, illustrating how infection can serve as an environmental "second hit" (Finding 6).

#### 6. Mechanism / Pathophysiology

**Causal chain (upstream → downstream):**



**Molecular pathway / cellular processes:** Hemostasis — VWF-dependent platelet adhesion and aggregation (GO:0007596 blood coagulation; GO:0070527 platelet aggregation). Proteolysis (GO:0006508) by the metalloprotease.

**Protein dysfunction:** Loss of function via impaired secretion, misfolding, truncation, or reduced catalytic activity. The wild-type enzyme is regulated by a **closed→open allosteric conformational switch** (Spacer–CUB autoinhibition relieved by VWF-D4 binding, ~2-fold kcat increase; Finding 8). Missense variants can disrupt secretion, catalysis, or allosteric regulation.

**Suggested GO/CL/CHEBI terms:** GO:0004222 (metalloendopeptidase activity), GO:0005576 (extracellular region); CL:0000232 (erythrocyte), CL:0000233 (platelet), CL:0002679 (hepatic stellate cell), CL:0000115 (endothelial cell); CHEBI:29108 (calcium, cofactor), CHEBI:18420 (zinc, catalytic metal).

## 7. Anatomical Structures Affected

| Level             | Structures                                                                                   | Suggested ontology                                                   |
|-------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Organ (primary)   | Brain, kidney, heart                                                                         | UBERON:0000955,<br>UBERON:0002113,<br>UBERON:0000948                 |
| Organ (secondary) | Spleen, GI tract, placenta                                                                   | UBERON:0002106,<br>UBERON:0000160,<br>UBERON:0001987                 |
| Body systems      | Cardiovascular/hematologic; nervous; renal                                                   | —                                                                    |
| Tissue/cell       | Microvascular endothelium; platelets; erythrocytes; hepatic stellate cells (ADAMTS13 source) | CL:0000115; CL:0000233;<br>CL:0000232; CL:0002679                    |
| Subcellular       | Endothelial Weibel–Palade bodies (VWF storage); extracellular plasma compartment             | GO:0033093 (Weibel–Palade body);<br>GO:0005615 (extracellular space) |
| Localization      | Systemic microvasculature; bilateral/diffuse                                                 | —                                                                    |

Damage is **bilateral and diffuse** (microvascular), not focal or lateralized.

## 8. Temporal Development

**Onset:** Congenital deficiency present from birth; clinical onset ranges from **neonatal (35–50% present in first days of life with hemolysis, jaundice, thrombocytopenia)** through childhood to **adult/pregnancy-triggered late onset** (Findings 5, 10). Onset pattern of acute episodes is **acute/subacute**.

**Progression:** The disease course is **episodic/relapsing-remitting** overlaid on a **chronic, lifelong** deficiency state. Acute events cluster around triggers. Progression rate is variable and partly determined by residual ADAMTS13 activity — higher residual activity correlates with later overt onset (Findings 5, 10).

**Patterns:** Remission is **treatment-induced** (prophylactic replacement) between episodes; there is no spontaneous cure. Critical vulnerability windows are the **neonatal period** and **pregnancy**, which are also the key windows for prophylactic intervention.

## 9. Inheritance and Population

**Epidemiology:** Ultra-rare. Estimated prevalence on the order of ~1–2 per million (cTTP accounts for a minority of all TTP; overall TTP incidence ~1.5–6 per million/year). Orphanet classifies it among ultra-rare disorders.

**Genetic parameters:** - **Inheritance:** Autosomal recessive (Finding 1). - **Penetrance:** High at the biochemical level (biallelic LoF → severe deficiency), but **clinical penetrance/expressivity is variable and incomplete** — the registry documents heterogeneous course and imperfect genotype-phenotype correlation (PMID: 30792199). - **Expressivity:** Variable, modulated by residual activity, modifier genes, and triggers. - **Founder effects:** c.4143dupA in Northern/Central Europe (Finding 4). - **Consanguinity:** Increases homozygous cases in populations with high consanguinity. - **Genetic anticipation / mosaicism:** Not applicable / not established.

**Demographics:** Marked **female predominance (~78%)**, driven by pregnancy-triggered presentation (Finding 5). Affects all ethnicities; specific founder variants show geographic clustering (European c.4143dupA). Age distribution is bimodal-ish: neonatal peak and a young-adult female (pregnancy) peak.

## 10. Diagnostics

**Core laboratory diagnosis:** Severe **ADAMTS13 activity <10%** measured by the **gold-standard FRETs-VWF73 fluorogenic assay** (Finding 9). Supportive labs: thrombocytopenia, MAHA with schistocytes on smear, elevated LDH, low haptoglobin, elevated indirect bilirubin, negative direct antiglobulin test.

**Distinguishing cTTP from iTTP:** In cTTP, an **anti-ADAMTS13 inhibitor/autoantibody is ABSENT** and **biallelic pathogenic ADAMTS13 variants are present**. In acquired iTTP, an inhibitor is present (Finding 9). Mixing studies and anti-ADAMTS13 IgG assays help discriminate.

**Genetic testing:** **ADAMTS13 single-gene sequencing** (or gene panels including complement genes for TMA differential; PMID: 30046676) is definitive. Genetic analysis is recommended in TTP patients **of all ages once an inhibitor is excluded** (Finding 10). WES/WGS are useful when panels are non-diagnostic. Chromosomal microarray/karyotype/FISH are not indicated (single-gene disorder).

**Emerging assays:** HemosIL AcuStar (faster, but reads systematically lower, with diagnostic discrepancies) and fiber-optic SPR immunoassays (Finding 9).

**Differential diagnosis:** Acquired iTTP; atypical HUS (complement dysregulation); Shiga-toxin HUS; DIC; HELLP/pregnancy TMA; Evans syndrome; immune thrombocytopenia (ITP) — the latter is a recognized misdiagnosis before hTTP is revealed by pregnancy loss (PMID: 39614241).

## 11. Outcome/Prognosis

**Mortality/morbidity:** Untreated acute events are **life-threatening**; historically high mortality. The dominant modifiable determinant of outcome is prophylaxis — **87% of acute events occurred without prophylaxis, and 20% of those caused organ damage** (Finding 5). Recurrent strokes and chronic kidney damage drive long-term morbidity and disability.

**Life expectancy:** With appropriate ADAMTS13 replacement prophylaxis, acute events are largely preventable and long-term outlook is substantially improved; without it, recurrent life-threatening episodes and cumulative ischemic organ damage occur.

**Pregnancy outcomes:** Untreated pregnancies carry high risk — 24% loss and 45% severe obstetric morbidity in one cohort; plasma prophylaxis reduces SOM to 28% (from 72%) (Finding 7).

**Prognostic factors/biomarkers:** Residual ADAMTS13 activity (higher = later/milder onset; Findings 5, 10); adherence to prophylaxis; and **elevated non-pregnant VWF antigen predicts severe obstetric morbidity** (Finding 7) — a candidate prognostic biomarker.

## 12. Treatment

**Principle: ADAMTS13 replacement.**

| Modality                             | Agent                                                   | Notes                                                                         | Suggested NCIT             |
|--------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------|
| Recombinant ADAMTS13                 | Apadamtase alfa (Adzynma/TAK-755, BAX 930), 40 IU/kg IV | Prevented ~all acute events in phase 3 (NCT03393975); no anti-drug antibodies | NCIT: recombinant ADAMTS13 |
| Fresh frozen plasma                  | FFP infusion / plasma exchange                          | Historical mainstay; provides exogenous ADAMTS13                              | NCIT:C171453 (plasma)      |
| Plasma-derived FVIII/VWF concentrate | Koate (contains ADAMTS13)                               | Alternative source of enzyme; long-term data available (PMID: 37855744)       | —                          |

**Recombinant ADAMTS13** is now the transformative therapy (Finding 3): 40 IU/kg IV, prophylaxis prevented essentially all acute events with a favorable safety profile and no anti-ADAMTS13 antibody development. It also rescued a plasma-refractory pregnancy (Finding 7).

**Emerging/experimental:** ADAMTS13 **gene therapy**; additional recombinant products; and novel VWF-activity inhibitors are under development (PMID: 42422077). A **constitutively active ADAMTS13 variant (Ala1144Val, "caADAMTS13")** with ~5-fold enhanced activity shows thrombolytic/anti-inflammatory efficacy in murine stroke models (PMID: 34780600).

**Supportive care:** Individualized infusion intervals guided by ADAMTS13 activity or, where testing is limited, surrogate markers (platelet count, LDH). Therapeutic plasma exchange can prolong intervals between administrations in some patients (PMID: 30394580).

### 13. Prevention

- **Primary prevention:** Not applicable (genetic disease); **genetic counseling** and carrier screening in affected families and consanguineous populations.
- **Secondary prevention:** Early diagnosis via ADAMTS13 testing and genetic confirmation; neonatal vigilance in known-carrier families.
- **Tertiary prevention (core strategy): Prophylactic ADAMTS13 replacement** to prevent acute events and organ damage — especially intensified during pregnancy and the peripartum period (Findings 5, 7). Regimen individualization (e.g., FFP every 10 days intensified to weekly in third trimester) enables successful full-term pregnancy even in resource-limited settings (PMID: 42529691).
- **Counseling:** Genetic counseling for family planning, prenatal/carrier testing, and pregnancy management planning.

### 14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *ADAMTS13* is conserved across vertebrates. Mouse *Adamts13* (NCBI Gene) is the principal experimental ortholog. Phylogenetic analysis across 264 vertebrates shows **allosteric regulation of ADAMTS13 is broadly conserved** across placental mammals, birds, and amphibians (PMID: 30700419).
- **Natural disease / veterinary relevance:** No well-established naturally occurring cTTP-equivalent is prominently documented in companion animals in the reviewed literature; the disease is chiefly modeled experimentally (see Section 15).
- **Evolutionary conservation:** The T7 and T8 distal domains are essential for allosteric activation, while T3–T6 are dispensable and have been repeatedly deleted across lineages,

indicating conserved core mechanism with flexible peripheral architecture ([PMID: 30700419](#)).

## 15. Model Organisms

**Principal model — Adamts13-knockout mouse (Motto et al. 2005):** - **Type:** Mammalian genetic knockout. - **Phenotype recapitulation:** Partial — mice are viable with normal survival and prolonged VWF-platelet-endothelial interactions but do **not** spontaneously develop full TTP on a standard background. On the **CASA/Rk high-VWF background**, a subset develop spontaneous thrombocytopenia; **shigatoxin challenge produces a syndrome closely resembling human TTP** (Finding 6). - **Value:** Demonstrates the essential **gene-environment ("two-hit") architecture** and the existence of TTP-modifying genes beyond VWF. - **Limitation:** Requires modifier background plus trigger; does not capture the full spontaneous human phenotype in isolation.

**Engineered variant models:** The **constitutively active caADAMTS13 (Ala1144Val)** used in murine stroke models illustrates therapeutic proof-of-concept for conformationally activated enzyme ([PMID: 34780600](#)).

**Applications:** Studying VWF-ADAMTS13 axis, trigger biology, thrombolytic/therapeutic testing, and gene-therapy development.

**Resources:** MGI (mouse *Adamts13*), IMPC/KOMP for knockout resources.

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## Mechanistic Model / Interpretation

The unifying model of cTTP is a **single-enzyme deficiency with an amplifying, trigger-dependent thrombotic cascade**. ADAMTS13 is the sole physiological regulator of VWF multimer size. Its congenital absence is necessary but often **not sufficient** for overt disease — the near-zero baseline activity creates a "primed" state in which the addition of high shear or elevated VWF (from pregnancy, birth, infection, or inflammation) tips the balance toward runaway VWF-platelet microthrombosis. This explains three otherwise puzzling clinical observations: (1) the marked female predominance (pregnancy is the archetypal trigger), (2) the neonatal-birth vulnerability window, and (3) the variable, episodic course despite a fixed genetic lesion.

The **allosteric biology of ADAMTS13** (closed-Spacer/CUB autoinhibition relieved by VWF-D4 binding) is not merely mechanistic detail — it directly informs therapeutics. Understanding that catalytic output is governed by a conformational switch enabled engineering of **constitutively active variants** with enhanced thrombolytic potency, and it clarifies why recombinant enzyme replacement so effectively restores the missing regulatory function.

The **genotype-phenotype relationship** is best understood as a continuum of residual activity: null/severe alleles (frameshift, nonsense, founder c.4143dupA) tend toward neonatal/childhood onset, while residual-activity missense alleles (R1060W) permit late, trigger-dependent (pregnancy) presentation. This is coherent with the registry finding that higher residual ADAMTS13 activity predicts later overt onset.

Therapeutically, the disease has moved from a reactive, plasma-based paradigm to **proactive recombinant ADAMTS13 prophylaxis**, which in the pivotal trial reduced acute events essentially to zero. The natural-history data — that 87% of acute events occur without prophylaxis — provide the quantitative rationale for lifelong prophylactic replacement.

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## Evidence Base

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| PMID                     | Title (abbreviated)                             | Supports                                        | Evidence type          |
|--------------------------|-------------------------------------------------|-------------------------------------------------|------------------------|
| <a href="#">12393505</a> | ADAMTS13 mutations in childhood TTP             | Biallelic AR inheritance (F1)                   | Human clinical/genetic |
| <a href="#">37895305</a> | Hereditary TTP review                           | Causal gene & enzyme deficiency (F1)            | Review                 |
| <a href="#">19180123</a> | Mechanisms of microvascular thrombosis          | ADAMTS13 source & shear-dependent cleavage (F2) | Review                 |
| <a href="#">17146059</a> | Exosite interactions / tension-induced cleavage | Tyr1605-Met1606 scissile bond (F2)              | In vitro               |
| <a href="#">38692292</a> | Recombinant ADAMTS13 phase 3 (NEJM)             | Zero acute events on rADAMTS13 (F3)             | Human RCT              |
| <a href="#">28912376</a> | Recombinant ADAMTS13 first-in-human             | Safety, no anti-drug antibodies (F3)            | Human phase 1          |
| <a href="#">16807643</a> | Common origin of 4143insA                       | European founder mutation (F4)                  | Human genetic          |
| <a href="#">42603082</a> | Natural history retrospective cohort            | Event burden, prophylaxis protection (F5)       | Human cohort           |
| <a href="#">38536644</a> | hTTP / ductus arteriosus & newborn survival     | Neonatal onset 35-50% (F5)                      | Review/clinical        |
| <a href="#">16200209</a> | Shigatoxin triggers TTP in ADAMTS13-KO mice     | Gene-environment model (F6)                     | Model organism         |
| <a href="#">36889591</a> | VWF antigen & pregnancy complications           | Pregnancy risk & prophylaxis benefit (F7)       | Human cohort           |
| <a href="#">36546627</a> | Recombinant ADAMTS13 for hTTP                   | Rescue of plasma-refractory pregnancy (F7)      | Human case             |
| <a href="#">32196558</a> | Antibodies conformationally activate ADAMTS13   | Closed→open allosteric switch (F8)              | In vitro/structural    |
| <a href="#">26391536</a> | Linker regions & flexibility                    | Conformational activation basis (F8)            | In vitro/structural    |
| <a href="#">37063760</a> |                                                 |                                                 | Methodological         |

| PMID                     | Title (abbreviated)                      | Supports                                            | Evidence type          |
|--------------------------|------------------------------------------|-----------------------------------------------------|------------------------|
|                          | ADAMTS13 activity testing platforms      | FRETs-VWF73 gold standard, assay discrepancies (F9) |                        |
| <a href="#">37711907</a> | Novel FO-SPR immunoassay                 | <10% diagnostic threshold (F9)                      | Methodological         |
| <a href="#">24994604</a> | Late-onset pregnancy-induced cTTP        | R1060W residual-activity genotype-phenotype (F10)   | Human clinical/genetic |
| <a href="#">30792199</a> | International Hereditary TTP Registry    | Heterogeneous course, incomplete correlation        | Registry               |
| <a href="#">30700419</a> | Phylogenetic/functional analysis         | Conserved allostery; essential T7/T8 domains        | Comparative            |
| <a href="#">34780600</a> | Constitutively active ADAMTS13 in stroke | Engineered therapeutic variant                      | Model organism         |
| <a href="#">42422077</a> | Update on treatment options              | Gene therapy & novel agents                         | Review                 |
| <a href="#">16160007</a> | Polymorphism modulation of ADAMTS13      | Modifier polymorphisms                              | In vitro               |

## Limitations and Knowledge Gaps

- 1. Epidemiology precision.** Exact prevalence/incidence of the congenital form is uncertain due to under-diagnosis and misclassification (e.g., misdiagnosis as ITP). Reported figures are estimates from registries and cohorts, not population-scale surveillance.
- 2. Incomplete genotype-phenotype correlation.** The registry explicitly documents heterogeneity that residual-activity alone does not fully explain; modifier genes beyond VWF are implicated by the mouse model but not fully mapped in humans.
- 3. Long-term outcomes of recombinant ADAMTS13.** The phase 3 trial establishes short/medium-term efficacy and safety; long-term (multi-decade) organ-protection, immunogenicity, and pregnancy-cohort data are still accruing.
- 4. Biomarker validation.** VWF antigen as a predictor of obstetric morbidity is promising but from a single small cohort and needs prospective validation.

5. **Animal model fidelity.** No single model spontaneously recapitulates the full human episodic phenotype without engineered modifier backgrounds and triggers.
6. **Gene therapy is preclinical/early.** Curative approaches remain investigational.

## Proposed Follow-up Experiments / Actions

1. **Prospective natural-history registry expansion** with standardized ADAMTS13 activity (FRETs-VWF73) and genotype capture to refine genotype–residual-activity–onset relationships and to validate VWF antigen as a prognostic biomarker.
2. **Long-term (5–10 year) follow-up of recombinant ADAMTS13** cohorts, including pregnancy sub-studies, immunogenicity surveillance, and organ-protection (renal/neurologic) endpoints.
3. **Modifier-gene mapping** in humans (GWAS/WGS in registry cohorts) to identify the TTP-modifying loci predicted by the CASA/Rk mouse studies.
4. **Advance ADAMTS13 gene therapy** from preclinical models toward first-in-human trials, leveraging the well-defined single-gene target and hepatic stellate-cell/hepatocyte expression biology.
5. **Standardize/harmonize ADAMTS13 activity assays** to resolve AcuStar vs FRETs-VWF73 discrepancies and validate rapid point-of-care assays (FO-SPR) for acute diagnosis.
6. **Structure-guided next-generation enzymes** — evaluate constitutively active (open-conformation) ADAMTS13 variants for acute event termination and potentially reduced dosing frequency.
7. **Pregnancy-management trials** comparing recombinant ADAMTS13 versus plasma prophylaxis regimens with maternal-fetal outcome endpoints.

*Report compiled from 10 confirmed findings across 5 investigation iterations and 31 reviewed papers. Evidence types span human RCT, human cohort/registry, human case reports, in vitro/structural, comparative/phylogenetic, and model-organism studies.*